Student Attendance Management System Using DevOps Tools

1. Introduction

The Student Attendance Management System is a web-based application designed to simplify daily attendance recording and monthly attendance calculation. Instead of entering attendance separately for each student, faculty members can view the entire class list on a single page and mark attendance for seven periods per day.

The system stores attendance records date-wise and period-wise. At the end of a selected month, it automatically calculates each student's total number of periods attended, periods absent, total working periods, and attendance percentage. This reduces manual calculations, saves faculty time, and helps identify students with low attendance.

The project also demonstrates the use of DevOps tools such as Git, GitHub, Jenkins, Docker, Kubernetes, and Selenium for version control, automation, containerization, deployment, and software testing.

2. Problem Statement

Manual attendance recording and monthly calculations require considerable time, particularly when a class contains many students and attendance must be recorded for multiple periods every day. Manual calculations can also lead to errors, difficulty retrieving past records, and delays in identifying students with low attendance.

The proposed system provides a centralized digital solution to record class-wise attendance across seven daily periods and automatically generate accurate monthly attendance summaries.

3. Objectives

1. Develop a web application to manage attendance for an entire class on one page.
2. Allow faculty to mark Present or Absent for each student across seven periods per day.
3. Store attendance records according to student, date, and period.
4. Calculate monthly working periods, attendance totals, absences, and attendance percentages automatically.
5. Generate individual student reports and class-wise monthly summaries.
6. Identify students whose attendance falls below a specified threshold.
7. Apply DevOps practices to manage, test, containerize, and deploy the application.

4. Functional Modules

Module 1: Student Management

- Add student roll number, name, and department.
- Display all registered students in a class list.
- Edit student details when required.

Module 2: Daily Attendance

- Select the attendance date.

- Display all students in a single table.
- Provide seven attendance columns representing Periods 1 to 7.
- Mark each period as Present or Absent.
- Provide Mark All Present and Mark All Absent options.
- Save and retrieve daily attendance records.

Module 3: Attendance Record Management

- Store attendance against the student, date, and period.
- Prevent duplicate entries for the same student and period.
- Allow authorized corrections to previously saved records.
- Retrieve attendance for a selected date or month.

Module 4: Monthly Attendance Calculation

- Calculate total working periods for the selected month.
- Calculate periods attended and periods absent for every student.
- Compute attendance percentage automatically.
- Display students below the configured attendance threshold.

Module 5: Reports

- Generate student-wise monthly attendance reports.
- Display class-wise attendance summaries.
- View daily attendance history.
- Export monthly reports as CSV for further use.

5. Technologies Used

- **Frontend:** HTML, CSS, and JavaScript.
- **Backend:** Node.js with Express.js.
- **Database:** SQLite or MySQL for persistent student and attendance records.
- **Version Control:** Git and GitHub.
- **Continuous Integration:** Jenkins.
- **Containerization:** Docker.
- **Orchestration and Deployment:** Kubernetes.
- **Automated Testing:** Selenium WebDriver with JavaScript and Chrome.

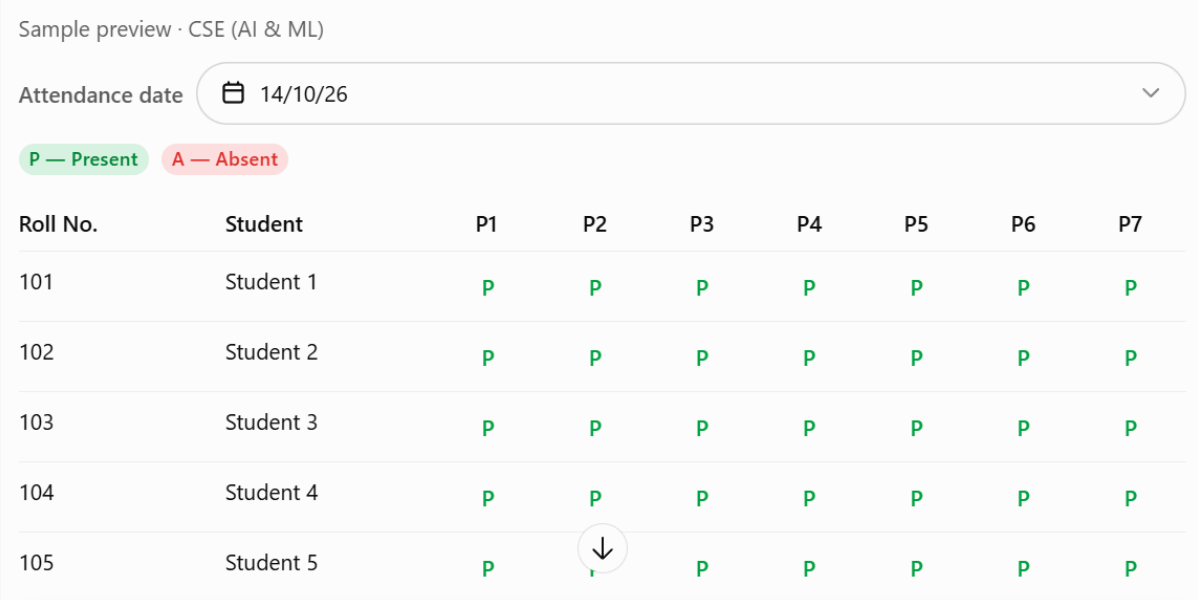
6. DevOps Implementation

1. Develop the attendance application and maintain its source code in GitHub.

2. Configure Jenkins to retrieve the source code and run automated validation or build steps.
3. Create a Docker image containing the application and its required dependencies.
4. Deploy the containerized application using Kubernetes Deployment and Service configurations.
5. Use Selenium to automate test cases for attendance entry, validation, monthly calculation, and report generation.
6. Demonstrate the deployed application and verify its functionality.

7. Selenium Test Cases

1. Verify that the attendance page displays all registered students.
2. Verify that attendance can be marked for all seven periods.
3. Verify that attendance is saved successfully for a selected date.
4. Verify that missing or invalid student information is handled correctly.
5. Verify that duplicate attendance entries are prevented.
6. Verify that monthly attendance totals are calculated correctly.
7. Verify that the attendance percentage is displayed correctly.
8. Verify that students below the configured attendance threshold are identified.
9. Verify that the monthly report can be generated and exported.
10. Verify that saved records remain available after the application is restarted.



Sample monthly report

Roll No.	Student	Working periods	Present	Absent	Attendance
101	Student 1	154	140	14	90.91%
102	Student 2	154	130	24	84.42%
103	Student 3	154	115	39	74.68%
104	Student 4	154	150	4	97.40%